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Economic Evaluation of Low-Vision Rehabilitation for Veterans With Macular Diseases in the US Department of Veterans Affairs

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IMPORTANCE Examining costs and consequences of different low-vision (LV) programs provides important information about resources needed to expand treatment options efficiently.

OBJECTIVE To examine the costs and consequences of LV rehabilitation or basic LV services.

DESIGN, SETTING, AND PARTICIPANTS The US Department of Veterans Affairs (VA) Low Vision Intervention Trial (LOVIT) II was conducted from September 27, 2010, to July 31, 2014, at 9 VA facilities and included 323 veterans with macular diseases and a best-corrected distance visual acuity of 20/50 to 20/200. Veterans were randomized to receive basic LV services that provided LV devices without therapy, or LV rehabilitation that added a therapist to LV services who provided instruction and homework on using LV devices, eccentric viewing, and environmental modification. We compared costs and consequences between these groups.

INTERVENTIONS Low-vision devices without therapy and LV devices with therapy.

MAIN OUTCOMES AND MEASURES Costs of providing basic LV services or LV rehabilitation were assessed. We measured consequences as changes in functional visual ability from baseline to follow-up 4 months after randomization using the VA Low Vision Visual Functioning Questionnaire. Visual ability was measured in dimensionless log odds units (logits).

RESULTS Of 323 randomized patients, the mean (SD) age was 80 (10.5) years, 314 (97.2%) were men, and 292 (90.4%) were white. One hundred sixty (49.5%) received basic LV services and 163 (50.1%) received LV rehabilitation. The mean (SD) total direct health care costs per patient were similar between patients who were randomized to receive basic LV services ($1662 [$671]) or LV rehabilitation ($1788 [$864]) (basic LV services, $126 lower; 95% CI, $299 lower to $35 higher; $P = .15$). However, basic LV services required less time and had lower transportation costs. Patients receiving LV rehabilitation had greater improvements in overall visual ability, reading ability, visual information processing, and visual motor skill scores.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT00958360.

Invited Commentary

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Chronic visual impairments (low vision [LV] and blindness) are among the 10 most prevalent causes of disability in the United States,[1] with an estimated 240 000 new cases of visual impairment annually.[2] Visual impairments often limit functional ability, hindering the performance of common tasks, such as reading, social interaction, and recreation.[3] The estimated financial burden of visual problems in the United States was more than $139 billion annually in 2013.[4] The leading causes of LV are diseases most prevalent in older patients, such as patients typically treated at US Department of Veterans Affairs (VA) health care facilities, including age-related macular degeneration, glaucoma, diabetic retinopathy, and optic neuropathies.[5] More than 1 million US veterans age 45 years and older have visual impairments: 157 000 with legal blindness and 1 026 000 with LV not severe enough to be classified as legally blind.[6]

Although no cures exist for most of the diseases causing age-related vision loss, rehabilitation programs have the potential to restore independence and improve quality of life for individuals with vision loss. The VA has established a comprehensive nationwide rehabilitation system called the "Continuum of Care for Visually Impaired Veterans," designed to enhance inpatient services provided by VA blind rehabilitation centers and expand outpatient LV and blind rehabilitation services.[6] Under this system, each of the VA's regional networks has LV services, including a basic LV service available at all VA eye clinics, as well as intermediate and advanced LV services at selected facilities. While previous studies have found that VA rehabilitation programs have a very large positive association with improved functional ability, few studies have compared the costs and consequences of different services along this continuum of care, which are needed to guide policy makers and develop informed clinical practice guidelines.[7]

The VA Low Vision Intervention Trial II (LOVIT II) study involved a multicenter, randomized clinical trial (RCT) at VA sites to compare LV rehabilitation with basic LV services. To our knowledge, LOVIT II was the first multicenter RCT comparing LV rehabilitation with basic LV services for patients with macular diseases and near normal or moderate levels of visual impairment.[8,9] In this study, we examined the costs and consequences of LV rehabilitation or basic LV services.

Key Points

Question What are the costs and consequences of low-vision rehabilitation (including therapy and homework to teaching low-vision device use, eccentric viewing, and environmental modification) compared with basic low-vision services (low-vision devices dispensed without therapy) for veterans with macular diseases and a visual acuity of 20/50 to 20/200?

Findings In this randomized clinical trial, health care costs were similar between patients receiving low-vision rehabilitation and basic low-vision services. However, low-vision rehabilitation required more time and transportation but was more effective for some patients.

Meaning These findings suggest that low-vision rehabilitation was more effective with similar health care costs; however, low-vision rehabilitation may involve a greater time commitment and cost to patients.

Methods

Setting and Patients

The design of the LOVIT II study has been described elsewhere.[8,9] Briefly, 323 patients eligible for VA health care services with a primary eye diagnosis (better-seeing eye) of macular disease and a best-corrected distance visual acuity of 0.40 to 1.00 logMAR (6/15-6/60 or 20/50-20/200 Snellen), using the Early Treatment of Diabetic Retinopathy Chart were randomized to receive LV rehabilitation or basic LV services at 9 VA facilities. All participants received an optometric LV examination, and they were eligible to receive the same LV devices without charge.

With basic LV services, an optometrist performs an LV examination and provides LV devices with minimal instructions about their use and maintenance. Because most LV clinics in the study did not stock LV devices, patients with basic LV services may have had a second trip to the VA for device dispensing. With LV rehabilitation, both an optometrist and LV therapist are involved. An LV therapist dispensed the LV devices and provided 2 or 3 (1.5- to 2.5-hour) therapy sessions to improve the use of remaining vision and use of LV devices. Contact time with the therapist depended on the devices prescribed and the patient's progress learning the skills that were taught. Low-vision rehabilitation also included structured homework, allowing patients to practice the effective use of remaining vision and LV devices.

LOVIT II assessed changes in visual reading ability (estimated from patients' difficulty ratings of reading items on the VA Low Vision Visual Functioning Questionnaire [VA LV VFQ-48])[10-12] between the LV rehabilitation and basic LV service arms from preintervention baseline to 4 months (2 months after completion of treatment). Additional measures included changes in overall visual ability, mobility, visual information processing, and visual motor skills.

This study was conducted in compliance with the tenets of the Declaration of Helsinki for research in human participants. The protocol and written informed consents were approved by the VA Central Institutional Review Board. Study oversight was provided by an independent data monitoring committee and the VA Cooperative Studies Program Coordinating Center (Hines, Illinois).

Study Design

We compared health care utilization, costs, and consequences between patients in the LOVIT II study who received either LV rehabilitation or basic LV services. We measured direct health care utilization and costs of LV rehabilitation and basic LV service from the health care clinician's (ie, the VA's) perspective. Costs were converted to 2017 US dollars using the Personal Consumption Expenditures health index, which is the Bureau of Economic Analysis's price index for health care consumption.[13] Additionally, we assessed resource utilization from the patients' and informal caregivers' perspectives.

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Informal caregivers were family members or friends of the patient. We measured consequences as changes in functional visual ability from baseline to 4 months after randomization using the VA LV VFQ-48.

Utilization and Cost Assessment

For basic LV services, direct health care use and costs were assessed for the initial optometry examination, a device-dispensing visit if needed, and the devices. For LV rehabilitation, direct utilization and costs were assessed for the initial optometry examination, therapy sessions, and devices. Low-vision devices were dispensed during therapy sessions for the LV rehabilitation group. We assessed personnel time from patient-level study forms or through staff interviews and applied relevant wage rates. The VA overhead costs were determined as a proportion of direct costs.[14] The costs of the LV devices were based on acquisition costs.

Indirect resource utilization and costs were assessed from the patient's and informal caregiver's perspectives. Indirect resource utilization consisted of time and transportation. We estimated the time that patients spent traveling to and receiving basic LV services or LV rehabilitation. For patients who received basic LV services, this consisted of the time traveling to and receiving the initial optometry examination and the time traveling to and receiving the LV devices. For patients who received LV rehabilitation, this consisted of the time traveling to and receiving the initial optometry examination, traveling to and receiving the LV therapy sessions, and the time performing the homework. We estimated the time traveling to health care facilities using web-based mapping software based on zone improvement plan codes of veterans and LV outpatient centers. Veterans kept logs of time spent on homework. Additionally, LV therapists recorded the time spent in each training session. Informal caregivers' time included time spent with veterans traveling to and receiving LV services and time assisting with activities of daily living. Informal caregivers' time spent assisting patients with activities of daily living during a typical week was collected during each study assessment. We also estimated distances from veterans' homes to VA LV outpatient centers and calculated transportation costs using federal reimbursement rates.[15]

Functional Visual Ability Assessment

Functional visual ability was assessed using the VA LV VFQ-48, which was developed to measure outcomes of vision rehabilitation programs.[10-12,16-18] The instrument contains 1 primary question with 3 subquestions for each of the 48 items. The items were chosen from 4 functional domains (reading ability, mobility, visual motor skills, and visual information processing) that depend heavily on vision and are affected positively by rehabilitation programs. The questionnaire was administered to patients by telephone before they participated in vision rehabilitation programs and at follow-up. Measures are estimated from a Rasch analysis and expressed in logits.[19] We calculated the change in logits from baseline to follow up for the overall VA LV VFQ-48 scores and each of the 4 functional domains.

Analysis

The analysis was performed on an intent-to-treat basis in which patients were analyzed based on the group to which they were randomized regardless of whether they crossed over to the other treatment group or left the study early. Of the 163 patients assigned to receive LV rehabilitation, 161 (98.8%) received the LV rehabilitation as assigned, and of the 160 patients assigned to receive basic LV services, 154 (96.3%) received the services as assigned.[20] We performed a cost-consequences analysis to compare costs and outcomes of the LV rehabilitation or basic LV services. We calculated 95% CIs around differences in mean utilization and costs between patients randomized to receive LV rehabilitation or basic LV services using bootstrapping procedures.[21] Differences were statistically significant at the 5% level if the 95% CIs did not include 0. We compared consequences in terms of functional visual ability between patients in the LV rehabilitation or basic LV services groups. We measured consequences as changes in the VA LV VFQ-48 visual ability measures between baseline and follow up. Differences in mean VA LV VFQ-48 visual ability measures between rehabilitation and basic LV services were compared using *t* tests. Cohen *d* was used to calculate the magnitude of treatment effects as small (0.2), medium (0.5), or large (0.8).[22] SAS software, version 9.4 (SAS Institute), was used to perform all analyses.

Sensitivity Analyses

As sensitivity analyses, we examined both costs and consequences by preplanned stratification of a best-corrected distance visual acuity better-eye (BCDVA better-eye) of 20/50 to 20/63 and worse than 20/63 to 20/200.[20] Additionally, the LV devices were often not stocked, requiring a second dispensing visit. We estimated indirect resource utilization and transportation cost savings to patients and informal caregivers that could occur if the devices were stocked and patients could receive their LV devices during the day of the initial LV examination.

Results

The mean (SD) age of patients in both groups was approximately 80 (10.5) years, 314 (97.2%) were men, and 292 (90.4%) were white. The study sample is described elsewhere.[20]

Resource Use and Costs

Although time for the initial optometry examination was similar for both groups (1.41 hours), patients with basic LV services often required a second device-dispensing visit (**Table 1**). Consequently, the average direct costs per patient to the VA for the LV optometry examination were $87 (95% CI, $76-$97; $P < .001$) higher for the basic LV services group. Most patients in the LV rehabilitation group had 3 visits (2 for therapy [mean, 1.9] and 1 for the LV examination). Low-vision therapists averaged over 6.3 hours per patient with a mean (SD) cost of $276 ($145) per patient. The total costs of LV devices were $178 higher in the basic LV group (95% CI, $21-$327 higher; $P = .02$). However, the mean (SD) total direct costs per patient of basic LV services ($1662 [$671]) and LV rehabilitation

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Table 1. Direct Health Care Utilization and Costs

Characteristic	Mean (SD)				Difference, Mean (95% CI)[a]	
	Basic LV Services (n = 160)		LV Rehabilitation (n = 163)			
	Resource Use, h	Cost, $[b]	Resource Use, h	Cost, $[b]	Resource Use, h	Cost, $
Optometry examination	NA	385 (55)	NA	298 (42)	NA	87 (76 to 97)
Optometrist's time/cost	3.45 (0.51)	261 (39)	2.60 (0.39)	197 (29)	0.85 (0.76 to 0.95)	65 (57 to 72)
Optometrist's preparation and completion time	1.17 (0.22)	88 (17)	1.17 (0.22)	89 (17)	−0.003 (−0.05 to 0.05)	0 (−4 to 3)
Examination	1.41 (0.38)	107 (29)	1.41 (0.36)	107 (28)	0.0016 (−0.076 to 0.081)	0 (−7 to 6)
Device dispensing with demonstration of use and maintenance	0.86 (0.26)	65 (20)	NA	NA	NA	NA
Additional time/cost	0.02 (0.11)	1 (8)	0.02 (0.13)	2 (10)	−0.0048 (−0.033 to 0.021)	0 (−2 to 2)
Administrative and clerical support for optometry examination	0.33 (0.02)	10 (1)	0.44 (0.06)	13 (2)	−0.12 (−0.13 to −0.10)	−3 (−4 to −3)
Overhead for optomety examination	NA	114 (16)	NA	88 (12)	NA	26 (22 to 29)
LV therapy	NA	13 (8)	NA	404 (208)	NA	−391 (−424 to −360)
Therapist's time/cost	0.10 (0.13)	4 (6)	6.31 (3.32)	276 (145)	−6.21 (−6.69 to −5.69)	−272 (−295 to −249)
Therapist's preparation and completion time	NA	NA	2.39 (1.09)	105 (48)	NA	NA
Therapy session/homework review	NA	NA	3.74 (2.56)	164 (112)	NA	NA
Additional time/cost	0.10 (0.13)	4 (6)	0.17 (0.34)	8 (15)	−0.078 (−0.15 to −0.031)	−3 (−6 to −1)
Administrative and clerical support for LV therapy	0.16 (0.02)	5 (1)	0.28 (0.06)	8 (2)	−0.12 (−0.13 to −0.10)	−3 (−4 to −3)
Overhead for LV therapy	NA	4 (2)	NA	119 (61)	NA	−115 (−125 to −106)
Total optometry and therapy cost	NA	390 (82)	NA	702 (208)	NA	−311 (−347 to −278)
Devices and supplies	NA	1265 (655)	NA	1086 (741)	NA	178 (21 to 327)
Monocular telescope	NA	84 (92)	NA	63 (88)	NA	21 (1 to 40)
Teleloupes	NA	183 (211)	NA	150 (199)	NA	33 (−12 to 79)
Pocket magnifier	NA	39 (29)	NA	29 (27)	NA	11 (5 to 17)
Stand magnifier	NA	121 (107)	NA	100 (108)	NA	21 (−1 to 44)
Intermediate distance device	NA	8 (16)	NA	9 (17)	NA	−1 (−5 to 2)
Reading glasses	NA	95 (134)	NA	88 (131)	NA	7 (−22 to 36)
Large inline video magnifier	NA	388 (538)	NA	360 (528)	NA	28 (−89 to 144)
Portable video magnifier	NA	206 (282)	NA	145 (255)	NA	62 (3 to 121)
Outdoor filters	NA	21 (23)	NA	21 (23)	NA	0 (−5 to 5)
Prescription sunglasses	NA	50 (115)	NA	38 (102)	NA	12 (−11 to 34)
Indoor filters	NA	7 (16)	NA	5 (14)	NA	2 (−1 to 6)
Reading stand	NA	23 (31)	NA	28 (32)	NA	−5 (−11 to 2)
Lamp	NA	36 (53)	NA	49 (56)	NA	−13 (−24 to 0)
Total direct cost	4.04 (0.54)	1662 (671)	9.63 (3.37)	1788 (864)	−5.59 (−6.12 to −5.11)	−126 (−299 to 35)

Abbreviations: LV, low vision; NA, not available.

[a] Difference in mean resource use or costs between patients randomized to receive basic LV service or LV rehabilitation (ie, basic LV service–LV rehabilitation).

[b] The hourly wage rates (not including benefits) were estimated as $58.31 for the optometrist, $33.66 for the LV therapist, and $22.81 for administrative and clerical support.

($1788 [$864]) were similar for overall study patients (basic LV services, $126 lower; 95% CI, $299 lower to $35 higher; $P = .15$).

From the patient's perspective, basic LV services involved a mean (SD) time commitment of 5.93 (2.43) hours per patient while LV rehabilitation required 12.86 (6.79) hours (6.93 fewer hours for basic LV services; 95% CI, 5.84-8.13 fewer hours; $P < .001$) (**Table 2**). Additionally, informal caregivers spent less time assisting patients with basic LV services because they spent less time driving to clinic visits (2.25 fewer hours for basic LV services; 95% CI, 1.15-3.44 fewer hours; $P < .001$). Moreover, transportation costs from the patient and caregiver's perspectives were $31 (95% CI, $21-$47; $P < .001$) lower for patients in the basic LV services group.

Sensitivity analyses are shown in **Table 3**. For patients with BCDVA better-eye scores of 20/50 to 20/63, the mean (SD) total direct health care costs per patient of basic LV services ($1386 [$583]) and LV rehabilitation ($1542 [$640])

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Table 2. Indirect Resource Utilization and Costs of Patients and Caregivers

Time	Mean (SD)				Difference, Mean (95% CI)[a]	
	Basic LV Services (n = 160)		LV Rehabilitation (n = 163)			
	Resource Use, h	Cost, $[b]	Resource Use, h	Cost, $[b]	Resource Use, h	Cost, $
Patient's time						
Patient's transportation to outpatient clinic	3.58 (2.42)	NA	5.69 (3.71)	NA	−2.11 (−2.84 to −1.47)	NA
Patient's care from optometrist	2.27 (0.50)	NA	1.41 (0.36)	NA	0.86 (0.76 to 0.95)	NA
Patient's therapy from LV therapist	0.08 (0.04)	NA	3.86 (2.61)	NA	−3.77 (−4.20 to −3.39)	NA
Patient's time doing homework	NA	NA	1.90 (2.26)	NA	NA	NA
Total patient time	5.93 (2.43)	NA	12.86 (6.79)	NA	−6.93 (−8.13 to −5.84)	NA
Caregiver's time	3.64 (3.46)	NA	5.89 (6.48)	NA	−2.25 (−3.44 to −1.15)	NA
Transportation to outpatient clinic[b]	53.19 (45.42)	28 (24)	112.60 (151.28)	60 (81)	−59.41 (−87.98 to −9.65)	−32 (−47 to −21)

Abbreviations: LV, low vision; NA, not available.

[a] Difference in mean resource use or costs between patients randomized to receive basic LV service or LV rehabilitation (ie, basic LV service−LV rehabilitation).

[b] Resource use for transportation was measured in miles. We estimated distances from veterans' homes to US Department of Veterans Affairs LV outpatient centers and estimated transportation costs using the federal reimbursement rate of $0.535 per mile for private automobiles.[14]

Table 3. Sensitivity Analyses

Characteristic	Mean (SD)				Difference, Mean (95% CI)[a]	
	Basic LV Services (n = 160)		LV Rehabilitation (n = 163)			
	Resource Use, h	Cost, $	Resource Use, h	Cost, $	Resource Use	Cost, $
Stratified by BCDVA						
BCDVA better-eye score 20/50-20/63 (n = 132)	3.99 (0.58)	1386 (583)	8.57 (2.31)	1542 (640)	−4.59 (−5.16 to −4.00)	−156 (−362 to 59)
BCDVA better-eye score <20/63-20/200 (n = 191)	4.07 (0.50)	1856 (663)	10.35 (3.77)	1955 (955)	−6.28 (−7.00 to −5.48)	−99 (−339 to 130)
LV devices dispensed on same visit as LV examination in basic LV services						
Patient time	3.23 (1.31)	NA	12.86 (6.79)	NA	−9.63 (−10.73 to −8.61)	NA
Caregiver's time	2.05 (1.92)	NA	5.89 (6.48)	NA	−3.84 (−5.00 to −2.95)	NA
Transportation to outpatient clinic	27.18 (22.95)	15 (12)	112.60 (151.28)	60 (81)	−85.42 (−114.25 to −66.54)	−46 (−63 to −36)

Abbreviations: BCDVA, best-corrected distance visual acuity; LV, low vision; NA, not available.

[a] Difference in mean resource use or costs between patients randomized to basic LV service or LV rehabilitation (ie, basic LV service−LV rehabilitation).

were similar (basic LV services $156 lower; 95% CI, $362 lower to $59 higher; P = .15). Additionally, for patients with BCDVA better-eye scores of less than 20/63 to 20/200, the mean (SD) total direct health care costs per patient of basic LV services ($1856 [$663]) and LV rehabilitation ($1955 [$955]) were similar (basic LV services $99 lower; 95% CI, $339 lower to $130 higher; P = .15). If LV devices were dispensed on the same day as the LV examination for patients with basic LV, then the patient's time would be 9.63 (95% CI, 8.61-10.73) hours lower, the informal caregiver's time would be 3.84 (95% CI, 2.95-5.00) hours lower, and transportation costs would be $46 (95% CI, $36-$63) lower for patients in the basic LV services group than patients in the LV rehabilitation group.

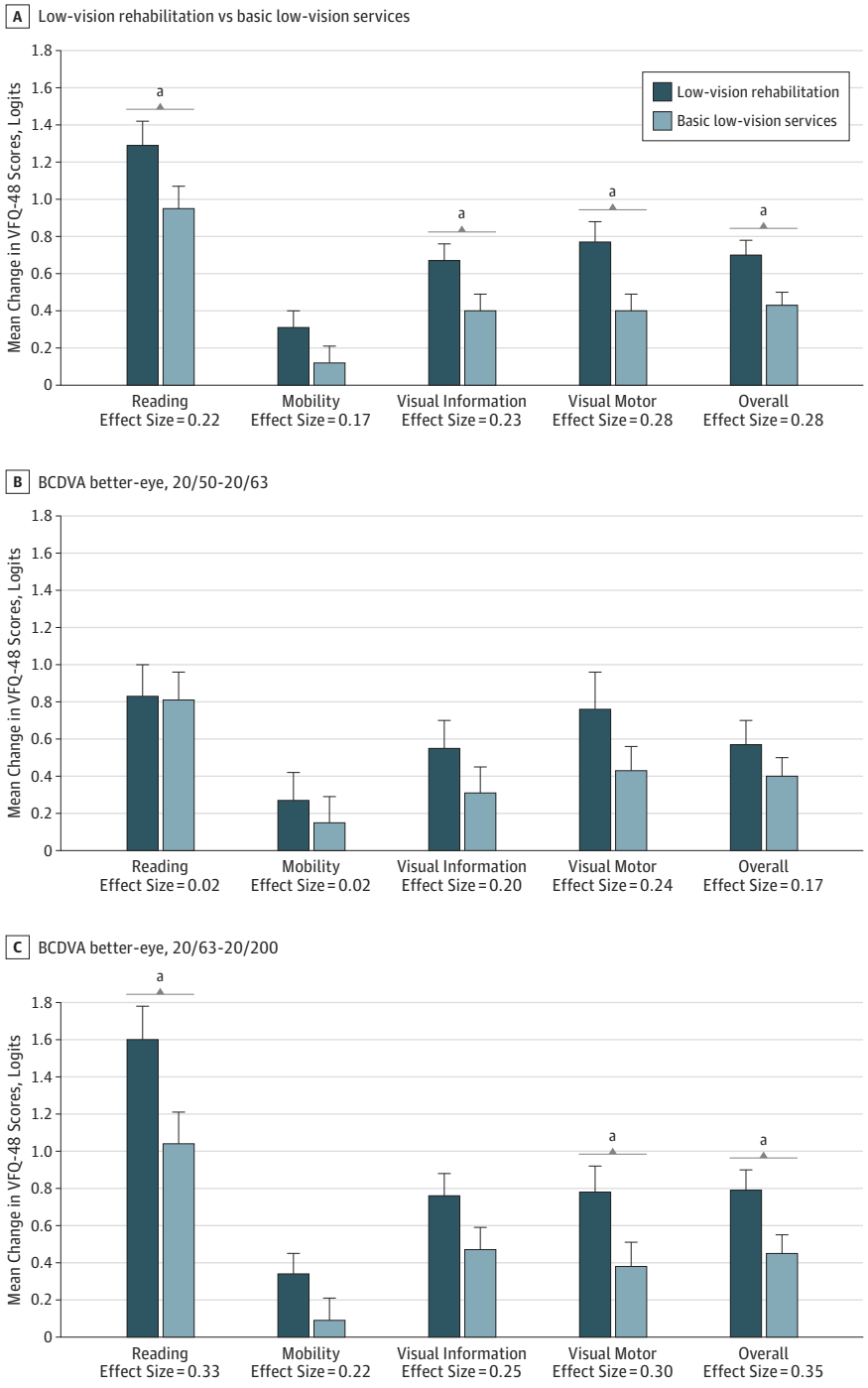
Functional Visual Ability

Changes in VA LV VHQ-48 scores between baseline and 4 months were greater for patients who received LV rehabilitation than for patients who received basic LV services for reading ability (0.34-logit greater improvement; 95% CI, 0.001-0.69; P= .05), overall visual ability (0.27-logit greater improvement; 95% CI, 0.06-0.49; P= .01), visual information processing (0.27-logit greater improvement; 95% CI, 0.01-0.53; P= .04), and visual motor skills (0.37-logit greater improvement; 95% CI, 0.08-0.66; P= .01) (**Figure**). The mean (SD) reading ability increased from 0.51 (1.43) to 1.80 (1.35) logits for the LV rehabilitation group, a 1.29- (1.66) logit increase, and from 0.51 (1.44) to 1.45 (1.68) logits for the basic LV group, a 0.95- (1.46) logit increase. Overall visual ability increased from a mean (SD) of 0.61 (1.10) to 1.32 (1.24) logits for patients receiving LV rehabilitation, a 0.70- (1.06) logit increase, and from 0.61 (1.02) to 1.04 (1.21) logits for patients with basic LV, a 0.43- (0.89) logit increase. However, when stratified by BCDVA better-eye, changes in VA LV VFQ-48 scores were only significant for patients with a BCDVA better-eye worse than 20/63 to 20/200.

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Figure. Comparison of Mean Changes in Veterans Affairs Low-Vision Visual Functioning Questionnaire (VA LV VFQ-48) Scores



A, Low-vision rehabilitation vs basic low-vision services. B, Low-vision rehabilitation vs basic low-vision services, best-corrected distance visual acuity (BCDVA) better-eye score of 20/50 to 20/63. C, Low-vision rehabilitation vs basic low-vision services, BCDVA better-eye score of more than 20/63 to 20/200. The effect size characterizes the magnitude of the treatment effect as small (0.2), medium (0.5), or large (0.8). The bars represent the standard error of the mean.

[a] $P \leq .05$.

Discussion

US Department of Veterans Affairs services for veterans with blindness and visual impairment include comprehensive inpatient rehabilitation programs as well as multidisciplinary and basic LV programs.[23] The LOVIT trial evaluated an intense outpatient LV rehabilitation program for veterans with legal blindness with macular diseases (habitual distance visual acuity better-eye worse than 20/100-20/400) compared with a waiting-list control group.[24] LOVIT II complemented LOVIT by comparing the outcomes of 2 types of outpatient LV programs for veterans who had less severe visual impairment from macular diseases (BCDVA better-eye, 20/50-20/200).[9,20] The mean (SD) direct health care costs were similar for basic LV services and LV rehabilitation: $1662 ($671) per patient for basic

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LV services compared with $1788 ($864) per patient for multidisciplinary LV rehabilitation. Moreover, LV rehabilitation was associated with improvement in several dimensions of functional visual ability, particularly for patients with greater impairment (ie, BCDVA better-eye worse than 20/63-20/200). However, LV rehabilitation may involve a greater time commitment and cost to patients and informal caregivers.

In the private sector, LV services are provided in outpatient settings.[25] Thus, our findings might be most generalizable to health care systems that provide a continuum of blind rehabilitation services or those that provide care limited to 1 setting that are considering an expansion of care delivery services. The transportation of veterans to the outpatient LV program may provide a barrier to access such a program. Multiple trips to an outpatient clinic may be an issue for some veterans. In sensitivity analyses, we showed that the time and transportation burden for patients and caregivers would be less if devices are stocked and patients could receive their LV devices on the same day as the examination.

The VA has committed over $40 million to establish a comprehensive nationwide rehabilitation system, including outpatient rehabilitation, for veterans and active-duty personnel with blindness or visual impairment. The 10 regional blind centers continue to provide comprehensive care, and outpatient rehabilitation capabilities have been developed.[26] LOVIT II provides evidence for the effectiveness of the basic and LV rehabilitation programs that are currently available in the system. Moreover, the current continuum of care programs could be refined based on the results from LOVIT II.

Limitations

Although our cost-consequences analysis cannot be compared with treatments for other conditions, it indicates to policy makers the costs associated with improving the functional visual ability of veterans with LV with macular diseases. Additionally, not all costs were collected at a patient level. Moreover, the costs included were those directly related to the intervention. The interventions' downstream effect on health care costs was not considered (eg, patients who were better equipped to deal with their LV may have fewer ophthalmology visits in the future and may avoid falls). Consequently, the results may understate the economic benefits of the intervention.

Conclusions

Low-vision rehabilitation was associated with improvement in several dimensions of visual function, with similar direct health care costs as for basic LV services. However, LV rehabilitation may involve a greater time commitment and costs to patients and caregivers. As the VA has committed additional resources to outpatient blind rehabilitation, the LOVIT and LOVIT II programs may provide a useful model for expanding outpatient LV services.

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